

ANALOG SINGLE AXIS SUN TRACKER WITH CMRR

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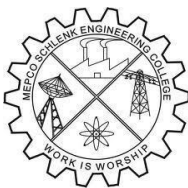
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CERTIFICATE

This is to certify that it is the Bonafide work done by Mr. NAVEEN PRAKASH. R (9517202403103), Mr. SIDDESHWARAN. V.P (9517202403154) for the 23EC353 - Mini Project - I work entitled Analog Single Axis Sun Tracker With CMRR at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025 – 2026.

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ABSTRACT

The demand for efficient solar energy harvesting is increasing rapidly, making intelligent tracking systems an essential part of modern photovoltaic setups. This project focuses on designing a **compact and reliable analog single-axis sun tracker** that continuously aligns a solar panel toward the maximum sunlight direction. The system uses a pair of Light Dependent Resistors (LDRs) as sensing elements, supported by a differential amplifier with an enhanced **Common-Mode Rejection Ratio (CMRR)** to accurately detect even small variations in illumination. By cutting down common-mode noise and ambient disturbances, the improved CMRR ensures stable tracking performance throughout the day.

The difference in light intensity between the sensors is processed through comparators and a motor-driving stage, allowing smooth and controlled rotation of the panel along one axis. The circuit is implemented using low-cost analog components, making the design lightweight, power-efficient and easy to fabricate. The system was tested under various lighting conditions and the results show consistent alignment accuracy and improved energy capture when compared to fixed-panel arrangements.

Overall, the proposed analog sun tracker provides a simple yet effective solution for maximizing solar panel output, making it suitable for educational demonstrations, small-scale renewable energy projects and environments where digital controllers are not preferred.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT BACKGROUND:

The increasing demand for renewable energy has made solar tracking systems essential for maximizing solar panel efficiency. Traditional fixed panels capture limited energy because they cannot follow the sun's movement throughout the day. To overcome this, a simple and cost-effective analog single-axis sun tracker is developed to automatically align the panel towards the sun using LDR-based sensing and a DC motor drive.

1.1.1 INTRODUCTION TO ANALOG SUN TRACKER:

With the growing demand for clean and renewable energy sources, solar power has become one of the most widely adopted technologies across the world. Although solar panels are capable of generating electricity from sunlight, their efficiency is heavily influenced by the angle at which sunlight strikes the surface. A fixed solar panel receives maximum energy only at a particular time of day, leading to reduced output during other hours. To overcome this limitation, sun-tracking systems have gained popularity in both industrial and small-scale energy applications. These systems allow the panel to continuously adjust its position to face the sun, ensuring improved power generation throughout the day.

Among the different types of tracking mechanisms, analog single-axis sun trackers stand out for their simplicity, low cost and ease of implementation. Unlike digital trackers that require microcontrollers and programmed logic, analog trackers rely on basic electronic components and natural light variations to determine the sun's direction. Light Dependent Resistors (LDRs) are commonly used as sensors due to their fast response and suitability for intensity comparison. By placing two LDRs in a specific orientation, even a slight difference in illumination produces a voltage imbalance that can be processed to determine the direction in which the panel should rotate.

In this project, a CMRR-enhanced differential amplifier plays a key role in improving the accuracy of the sensing system. A high Common-Mode Rejection Ratio (CMRR) ensures that unwanted noise, uniform lighting changes and environmental disturbances do not affect the tracking operation. This allows the tracker to respond only to the actual difference in sunlight falling on the two sensors, thereby increasing stability and precision. The processed signal is then passed to comparator and motor-driver stages, which controls the rotation of the panel through a DC motor.

Modern solar-tracking research focuses on achieving maximum efficiency with minimal complexity and cost. Many techniques have been reported, including dual-axis mechanisms, servo-based digital controllers and hybrid tracking algorithms. However, these systems often require complex circuitry, coding or additional calibration. The analog technique implemented in this project offers a practical alternative, especially for academic projects and small installations, where reliability, affordability and easy fabrication are more important than advanced automation.

Single-axis analog trackers are extremely useful in rural solar installations, student projects, mobile power units, and small-scale renewable setups. They significantly improve the energy output compared to fixed panels while keeping the design simple and requires low-maintenance. This project focuses on developing a compact, efficient and cost-effective analog tracking system that maintains the panel's alignment with the sun during daylight hours and ensures improved solar power generation with minimal circuitry.

1.1.2 DIFFERENT TYPES OF SUN TRACKER:

Sun tracking systems are broadly classified based on their movement mechanism and control strategy. The primary goal of any tracking system is to maintain the solar panel perpendicular to the sun's rays throughout the day, thereby maximizing energy capture. Over the years, various tracking methods have been developed to improve efficiency, reduce power loss, and simplify implementation. The major categories of sun-tracking systems are :

a) Single-Axis Sun Trackers

Single-axis trackers rotate the solar panel around one axis (either horizontal or vertical). In most small-scale and educational projects, the horizontal-axis configuration is preferred because it follows the sun's east-to-west motion across the sky. These trackers are simple in design, consume less power, and are cost-effective. Although they provide lower efficiency compared to dual-axis systems, their performance is significantly better than fixed solar panels, making them popular for compact and budget-friendly applications.

b) Dual-Axis Sun Trackers

Dual-axis trackers rotate along both horizontal and vertical axes, allowing them to follow the sun's position in both azimuth and elevation angles. This results in maximum energy capturing capability and highest efficiency among all tracking systems. However, they are mechanically complex, require more sensors and are relatively expensive. These trackers are typically used in large solar farms and advanced research setups where peak energy extraction is required.

c) Passive Sun Trackers

Passive trackers do not use electronic components or sensors. Instead, they rely on thermal expansion, fluids or mechanical balancing techniques to adjust the panel's position. The movement is triggered automatically by temperature differences caused by sunlight. While passive trackers eliminate the need for power and circuitry, they are less accurate and slower compared to active systems.

d) Active Sun Trackers

Active trackers use electrical, electronic or electromechanical components to detect the sun's position and adjust the panel accordingly. These systems typically include LDR sensors, differential amplifier stages, comparators and motor drivers. The accuracy of active trackers makes them highly suitable for practical solar installations. The project presented here is

also an example of an **analog active single-axis sun tracker which is enhanced with high CMRR**, ensuring stable and reliable tracking

e) Hybrid Sun Tracking Systems

Hybrid trackers combine active and passive mechanisms to achieve controlled, energy-efficient panel movement. They often use a combination of sensors and mechanical balancing to reduce motor usage and extend system life. Although less common, hybrid trackers offer a good balance between accuracy and power consumption.

1.1.3 ADVANTAGES OF SUN TRACKER:

- **Higher energy output** – Tracks the sun throughout the day, improving panel efficiency compared to fixed installations.
- **Better utilization of sunlight** – Ensures the panel always receives maximum possible illumination.
- **Simple and cost-effective** – Analog design avoids complex coding and reduces overall system cost.
- **Low maintenance** – Fewer electronic components make the system reliable and easy to maintain.
- **Improved performance stability** – High CMRR reduces noise and ensures accurate tracking even under changing light conditions.
- **Compact and lightweight** – Uses minimal components, making it easy to install in small-scale setups.
- **Energy efficient** – Motor operates only when adjustment is needed, conserving power.

1.1.4 LIMITATIONS OF SUN TRACKER:

- **Mechanical wear and tear** – Moving parts like motors may degrade over time.
- **Less stable in bad weather** – Wind, dust, or sudden light changes can affect tracking accuracy.
- **Requires proper alignment** – Sensors must be positioned correctly for

reliable operation.

- **Slightly higher setup cost** – More expensive than fixed panels due to motor and tracking components.
- **Limited movement range** – Single-axis systems cannot follow vertical sun movement, reducing efficiency compared to dual-axis trackers.
- **Periodic calibration needed** – Sensor sensitivity and motor alignment may need occasional adjustments. It offers lower Gain.

1.1.5 APPLICATIONS OF SUN TRACKER:

- **Solar power plants** – Improves energy production in small and large photovoltaic setups.
- **Rural and off-grid systems** – Provides higher efficiency where power availability is limited.
- **Portable solar units** – Useful for camping, field research and military portable power kits.
- **Educational and academic projects** – Ideal for learning renewable energy concepts and analog control systems.
- **Solar street lighting** – Helps maintain optimal alignment for maximum battery charging.
- **Agricultural solar pumps** – Enhances output for irrigation systems relying on solar energy.
- **Small home solar setups** – Boosts daily energy generation without complex digital electronics.

1.2 OBJECTIVE OF THE PROJECT:

- To design and implement an analog single-axis sun tracking system that continuously aligns the solar panel towards maximum sunlight.
- To improve tracking accuracy using a high-CMRR differential sensing circuit.
- To enhance solar power output while keeping the system simple, low-cost, and energy-efficient.

CHAPTER 2

DESIGN PARAMETERS AND PROCEDURES

2.1 DESIGN PARAMETERS:

The performance and accuracy of an analog single-axis sun tracking system depend on several important design parameters. These parameters ensure that the system reliably senses light variation, amplifies only the required differential signal and rotates the solar panel with controlled movement.

The key design parameters considered in this project are described in the following sub sections.

2.1.1 LIGHT SENSITIVITY RANGE:

The system uses Light Dependent Resistors (LDRs) as sensing elements. Their resistance varies inversely with light intensity. It is essential that the LDRs used have a suitable sensitivity range so that even small differences in sunlight levels between the two sensors can be detected effectively. Proper sensitivity ensures faster response, better direction accuracy, and smooth panel alignment.

2.1.2 DIFFERENTIAL VOLTAGE RESPONSE:

The core working principle is based on comparing two voltage signals generated from the LDR pair. The difference between these voltages determines the direction and magnitude of panel movement. The differential amplifier must be capable of producing a noticeable and stable output voltage difference even for small light variations.

2.1.3 COMMON MODE REJECTION RATIO (CMRR):

Since both sensors are exposed to the same environmental surroundings, a portion of the signal becomes common-mode noise (e.g., ambient intensity variations, electrical interference, shadow flicker). A high

CMRR is required to reject these unwanted signals so that the amplifier responds only to the actual difference in light intensity. This significantly improves accuracy, stability, and consistency of the tracking mechanism.

2.1.4 MOTOR ROTATION SPEED AND CONTROL:

The motor must rotate slowly and smoothly so that the solar panel does not oscillate or overshoot. A suitable motor driver and gear mechanism are required to regulate the speed and provide sufficient torque. The rotation must stop immediately when the intensity difference becomes zero to avoid unnecessary power consumption.

2.2 SENSOR SELECTION AND SIGNAL ACCURACY CONSIDERATION:

For an analog single-axis sun tracker, the choice of sensing elements and the quality of the received signal play a crucial role in achieving smooth and accurate tracking. Light Dependent Resistors (LDRs) are used in this project due to their simplicity, low power requirement, wide dynamic range and ability to directly interface with analog circuits without the need for complex signal conditioning. Two identical LDRs are placed with a small physical divider between them to create a measurable intensity difference whenever the sun is not perfectly aligned with the panel surface.

Apart from selecting suitable sensors, ensuring the accuracy of the electrical signal is equally important. Matching the LDR characteristics, maintaining symmetrical placement, shielding the sensors from stray reflections, and using a differential amplifier with high Common-Mode Rejection Ratio (CMRR) helps in reducing noise and environmental distortions. These combined factors contribute to precise angular movement of the motor and prevent unnecessary oscillations, resulting in efficient tracking and better energy-harvesting performance.

CHAPTER 3

PROPOSED SYSYTEM DESIGN

3.1 SYSTEM ARCHITECTURE AND FUNCTIONAL BLOCK

EXPLANATION:

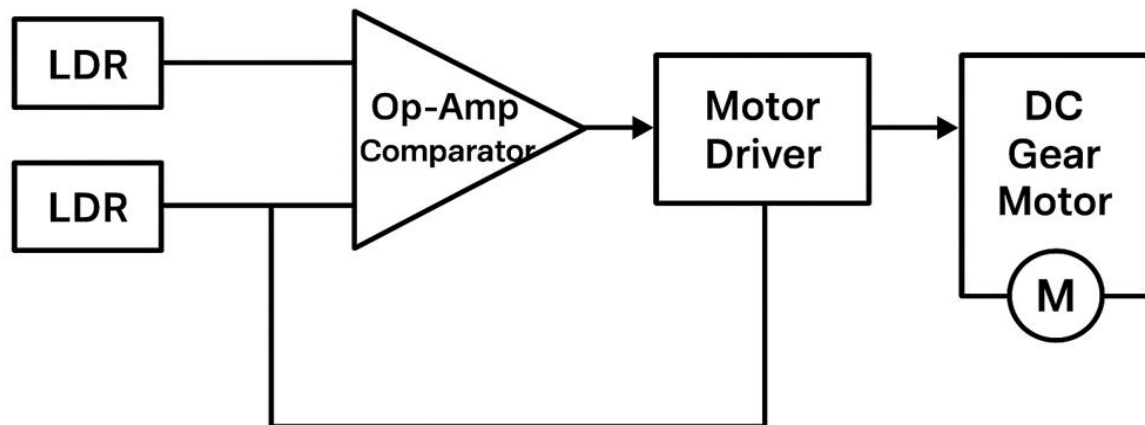


Fig 3.1 Block Diagram of Analog Sun Tracker

The analog single-axis sun tracker as shown in Fig 3.1 is designed using a simple yet effective block-based approach that ensures continuous alignment of the solar panel with the direction of maximum sunlight. The overall system operation is divided into four major functional blocks: the sensing block, signal conditioning block, motor control block and mechanical rotation block. Each block works together in real time to monitor the variation in light intensity and orient the panel accordingly.

The system begins with the **light sensing block**, where two Light-Dependent Resistors (LDRs) are positioned at slightly different angles to detect the difference in solar intensity from the left and right sides of the tracker. Any imbalance between the LDR outputs is passed on to the **signal conditioning block**, which is implemented using a high Common-Mode

Rejection Ratio (CMRR) differential amplifier or comparator stage. This stage converts the light variation into a clean, noise-free control signal.

Based on this processed signal, the **motor control block** activates a DC motor in the appropriate direction, ensuring smooth and controlled rotation. The **mechanical rotation block** consists of the mounting frame and gear mechanism which physically adjusts the solar panel until both LDRs receive equal illumination, indicating perfect alignment with the sun.

Through this continuous feedback mechanism, the system maintains accurate sun tracking without requiring digital processing, microcontrollers, or complex programming, making it efficient, low-cost and suitable for small-scale renewable energy applications.

3.1.1 FUNCTIONAL BLOCK DIAGRAM:

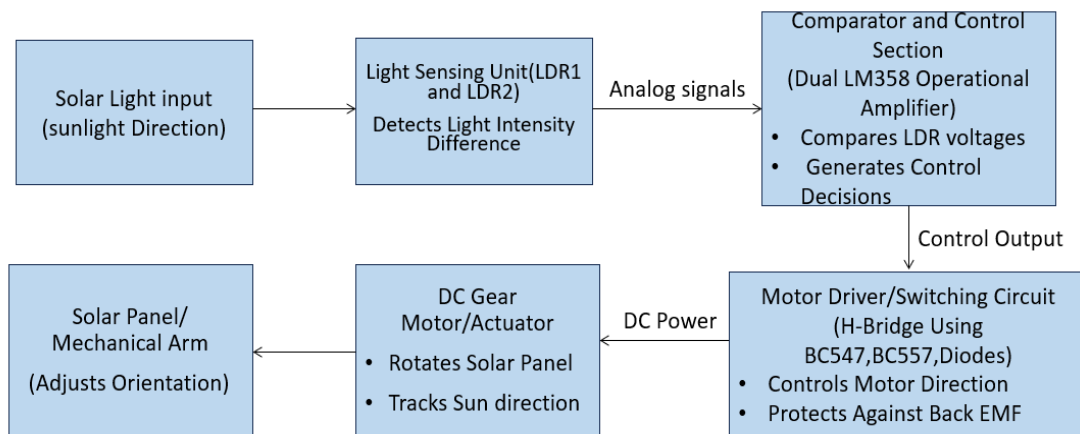


Fig 3.2 Functional Block Diagram of Analog Sun Tracker

The analog single-axis sun tracker uses two LDR sensors to detect the variation in sunlight intensity between left and right directions. These sensor outputs are fed into a dual LM358 comparator, which generates an appropriate control signal based on intensity difference. The signal drives an H-Bridge transistor circuit that rotates the DC motor either clockwise or anticlockwise. The mechanical assembly then positions the solar panel

towards the direction of maximum sunlight, ensuring continuous optimal alignment.

3.2 CIRCUIT EXPLANATION:

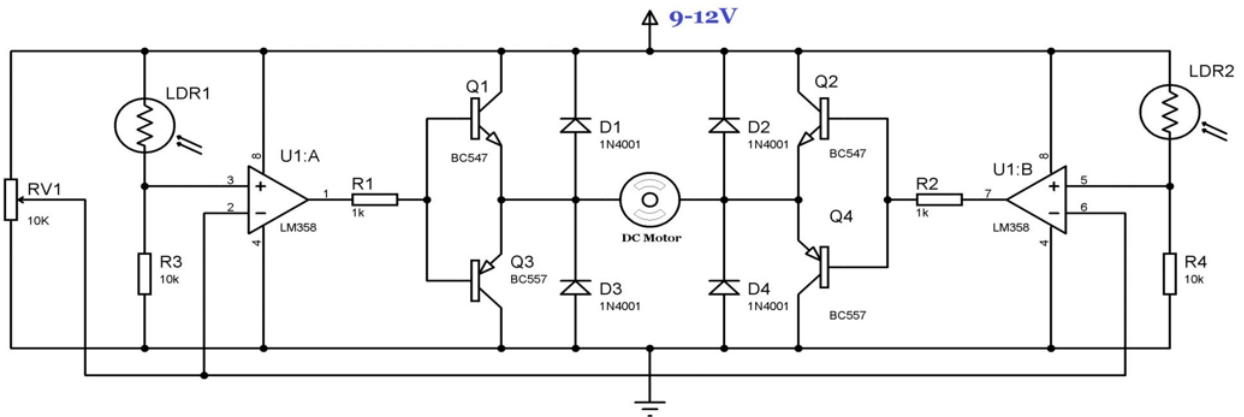


Fig 3.3 Circuit Diagram of Analog Single Axis Sun Tracker

The circuit consists of four stages and each shall be explained in sub sections 3.2.1 to 3.2.4

3.2.1 Light Sensing Stage:

Two Light Dependent Resistors (**LDR1** and **LDR2**) are positioned with a small divider wall between them to detect the difference in sunlight intensity. Their resistance decreases when exposed to higher light intensity and increases when light reduces. This changing resistance creates a corresponding voltage difference across the voltage divider networks formed using:

- **LDR1 + RV1 (10 k Ω potentiometer)**
- **LDR2 + R4 (10 k Ω resistor)**

The potentiometer (RV1) allows **manual calibration**, ensuring both sensors produce equal voltage under straight sunlight conditions

3.2.2 Differential Comparator Stage (LM358):

The LM358 IC contains two independent operational amplifiers (**U1: A** and **U1: B**). Each amplifier compares the sensor voltages and decides whether the motor needs to rotate **left**, **right**, or **stop**.

U1: A compares the voltage from LDR1 with a reference level determined by RV1

U1: B compares the voltage from LDR2 with a similar reference

Depending on which side receives more light, **one comparator output goes high** while the other remains low, generating a control signal for the H-bridge. As LM358 works well on a **single supply voltage** and has good **CMRR**, it accurately detects minor illumination differences without digital processing.

3.2.3 Motor Driver Stage (H-Bridge Configuration):

The motor driver uses **four Bipolar Junction Transistors (BJTs)** arranged in an H-bridge:

Q1 (BC547) and **Q3 (BC557)** form one half-bridge

Q2 (BC547) and **Q4 (BC557)** form the opposite side

The comparator outputs drive the transistor pairs as given in Table 3.1:

Table 3.1 Transistor pair and Motor Rotation Table

Comparator Output	Active Transistor Pair	Motor Rotation
U1:A HIGH	Q1 & Q4	Clockwise
U1:B HIGH	Q2 & Q3	Anti-clockwise
Both LOW or equal	None	Motor Stops

3.2.4 Protection Diodes:

To protect the transistors and IC from **back-EMF generated by the DC motor, four 1N4001 diodes (D1 - D4)** are placed across the H-bridge transistors. These diodes clamp the reverse voltage spikes and increase circuit reliability.

3.3 Mathematical justification — Analog Single-Axis Sun Tracker:

The mathematical Justification is provided for the circuit to verify that the chosen values for resistors and potentiometer are appropriate according to the requirement of the circuit

3.3.1 Sensor voltage (LDR voltage divider):

Each LDR is used in a simple voltage divider with a fixed resistor ($R = 10 \text{ k}\Omega$). If an LDR has resistance R_L and the lower resistor is R_b , the sensor node voltage (measured at the non-inverting or inverting input) is:

$$V_L = V_{CC} \frac{R_b}{R_L + R_b}$$

Where V_{CC} is the supply (usually 12V). This gives a monotonic relationship: when R_L decreases (brighter light), V_L increases.

Sensitivity (how much node voltage changes per ohm change in LDR):

$$\frac{dV_L}{dR_L} = -V_{CC} \frac{R_b}{(R_L + R_b)^2}$$

Example:

Take $V_{CC} = 12\text{V}$, $R_b = 10 \text{ k}\Omega$.

Bright LDR:

$$R_L = 1 \text{ k}\Omega \rightarrow$$

$$V_L = 12 \cdot \frac{10k}{1k+10k} = 12 \cdot \frac{10}{11} \approx 10.91 \text{ V.}$$

Dim LDR:

$$R_L = 100 \text{ k}\Omega \rightarrow$$

$$V_L = 12 \cdot \frac{10k}{100k+10k} = 12 \cdot \frac{10}{110} \approx 1.09 \text{ V.}$$

If $R_L \approx 10 \text{ k}\Omega$ (middle range) then:

$$\begin{aligned} \frac{dV_L}{dR_L} &= -12 \cdot \frac{10k}{(10k + 10k)^2} = -12 \cdot \frac{10k}{(20k)^2} = -\frac{120000}{400000000} \\ &= -3 \times 10^{-4} \text{ V}/\Omega \end{aligned}$$

So, a 1 k Ω change produces $\approx 0.3 \text{ V}$ change — easily detectable by the op-amp.

3.3.2 Differential input and comparator action (LM358):

The two sensor voltages are V_{L1} and V_{L2} . The op-amp (used open-loop / as comparator) reacts to the differential voltage:

$$V_{diff} = V_+ - V_- = V_{L1} - V_{L2}$$

Ideal op-amp behaviour (before saturation):

$$V_{out} \approx A_d \cdot V_{diff}$$

where A_d is the open-loop differential gain (very large). In practice the LM358 saturates near the rails:

$$V_{out} \approx \begin{cases} V_{sat+} & \text{if } V_{diff} > V_{th} \\ V_{sat-} & \text{if } V_{diff} < -V_{th} \end{cases}$$

$V_{sat+} \approx V_{CC} - 1.2 \text{ to } V_{CC} - 1.5 \text{ V}$ (LM358 single-supply)

$V_{sat-} \approx 0 \text{ V}$ (or a few tens of mV)

V_{th} is the effective switching threshold ($\approx$ input offset + small input noise). For LM358 typical input offset ~ 2 mV, so very small V_{diff} is sufficient.

Interpretation: even a small light imbalance gives a differential voltage that flips the comparator output to rail — hence clear left / right decision.

3.3.3 Role and math of CMRR :

Real sensor voltages contain two parts:

- Common-mode component $V_{CM} = \frac{V_{L1} + V_{L2}}{2}$ (changes due to global lighting, temperature, reflections)
- Differential component $V_{diff} = V_{L1} - V_{L2}$

An op-amp has finite ability to reject the common-mode voltage; this is quantified by **CMRR**:

$$\text{CMRR} = \frac{A_d}{A_{cm}} \text{ (linear) or } 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right) \text{ dB}$$

Where A_{cm} is the common-mode gain (ideally zero). A high CMRR means changes in V_{CM} produce little change at the output — i.e., clouds or overall brightness shifts will not trigger the motor. LM358 has decent CMRR (tens of dB), and choosing symmetrical divider resistances and matched LDRs reduces V_{CM} errors.

Practical effect: If a cloud reduces both sensor voltages by 1V, a high-CMRR stage will keep the output near previous state since $V_{diff} \approx 0$.

3.3.4 Output driving and transistor base current:

The comparator output (saturated high $\approx$ e.g. 10–11 V on a 12 V supply) is fed to base resistors $R1/R2 = 1$ k Ω before driving transistors (BC547/BC557). Approximate base current when op-amp output goes high:

$$I_B \approx \frac{V_{out} - V_{BE}}{R_1}$$

Take $V_{out} = 10.5\text{V}$, $V_{BE} \approx 0.7\text{V}$:

$$I_B \approx \frac{10.5 - 0.7}{1 \text{ k}\Omega} = \frac{9.8}{1000} = 9.8 \text{ mA}$$

With transistor DC current gain β (conservative $\beta = 50$ for saturation safe design; BC547 often 100 in active region but lower in saturation), collector current estimates to:

$$I_C \approx \beta \cdot I_B \approx 50 \cdot 9.8 \text{ mA} \approx 490 \text{ mA}$$

This is more than enough to drive a small DC geared motor (typical small motors draw 100–300 mA steady; stall currents can be higher — design for stall current with protective diodes already present).

Note: As complementary transistors are used in H-bridge halves, the combination provides current path for clockwise and counterclockwise rotation depending on which comparator output is high.

3.3.5 Minimum detectable illumination difference:

The sensitivity formula is combined with comparator switching as:

- Minimum $V_{diff,min}$ needed $\approx$ input offset voltage V_{os} (typically 1–5 mV for LM358) plus noise margin (conservative practical threshold is 10 mV).
- As sensitivity is dV/dR , the minimum LDR resistance difference $\Delta R_{L,min}$ the circuit can detect approximately:

$$\Delta R_{L,min} \approx \frac{V_{diff,min}}{\left| \frac{dV_L}{dR_L} \right|}$$

3.3.6 Importance of diodes (D1 – D4):

Diodes clamp the reverse voltage from inductive motor spikes. Back-EMF can be approximated as:

$$V_{emf} = K_e \cdot \omega$$

If not clamped, this can exceed transistor Vce and cause damage. The diodes

provide a safe conduction path, limiting negative spikes to about $-0.7V$ relative to the rail.

3.3.7 Summary of Mathematical Justification:

- The LDR divider gives a predictable voltage range ($\approx 1-11 V$ for typical LDRs with 12 V supply).
- The differential amplifier / comparator converts tiny voltage differences into saturated outputs that clearly decide motor direction.
- CMRR and matched resistor / sensor values ensure that common environmental changes don't trigger motion.
- With $R1 = 1 k\Omega$, the driver transistors see base currents sufficient for several hundred mA collector current — enough for typical small DC gear motors (gearbox recommended to reduce stall current).
- The circuit can detect very small LDR resistance changes (tens of ohms) which yields fine angular resolution, assuming good mechanical sensor mounting.

3.4 DESIGN CONSIDERATIONS:

When designing the analog single-axis sun tracker, the following practical considerations were observed to ensure reliable operation, safety and manufacturability.

• Sensor Matching & Placement

Two LDRs with similar characteristics are to be used, and symmetric placement with a small opaque divider is to be ensured so that differential sensitivity is maximized and systematic bias is reduced.

• CMRR & Signal Conditioning

An op-amp with good Common-Mode Rejection Ratio and low input offset is to be chosen, divider resistances are to be matched and filtering through small decoupling capacitors is to be applied to suppress flicker and fast transients.

• Hysteresis to Prevent Hunting

A small hysteresis (dead-band) at the comparator stage is to be added through slight positive feedback or threshold adjustment so that rapid motor oscillations near zero illumination difference are prevented.

- Motor Selection & Gearing

A DC gear motor with adequate stall torque is to be selected or gearing is to be employed so that motor load is reduced. Through gearing, current drawn is reduced and positional stability under wind loads is improved.

- Protection & Transient Suppression

Flyback diodes (already present) are to be used and an RC snubber or TVS across the motor is to be considered so that transistors and the op-amp are protected from inductive spikes.

- Power Budgeting & Efficiency

Low-power op-amps are to be chosen and the motor is to be operated only when correction is required so that harvested energy is not wasted and overall power consumption is minimized.

- Mechanical Rigidity & Bearings

A stiff mounting frame with smooth bearings is to be constructed so that friction, backlash and vibration are minimized, preventing degradation of positional accuracy and unnecessary increase in motor current.

- Environmental Robustness

The sensor mount and electronics are to be weatherproofed (with an IP-rated enclosure if outdoors). LDRs and wiring are to be protected from dust, moisture and water ingress while unobstructed light exposure is maintained.

- Calibration & Adjustment

A trim potentiometer (RV1) or other adjustment provision is to be included so the system can be zeroed during installation and periodically re-calibrated.

- Sensor Field of View & Shading

The sensor enclosure or tunnel is to be designed to limit stray light, reflections and shading from nearby structures is to be avoided so that misleading shadows are eliminated.

- Thermal Effects

Temperature-dependent component drift (op-amp offset, LDR

behaviour) is to be accounted for components rated for expected temperatures are to be used and temperature compensation should be added wherever required.

- **Safety & Fail-Safe Behaviour**

A fail-safe mechanism is to be implemented so that in case of sensor failure or power loss, a safe position (e.g., horizontal) is assumed or motor operation is stopped to prevent damage.

- **EMI/PCB Layout**

Analog input traces are to be kept short and sensor lines are to be separated from high-current paths and decoupling capacitors near op-amp power pins are to be added for EMI immunity.

- **Testing & Validation Plan**

Test points for sensor voltages, comparator outputs and motor drive currents are to be provided and system behavior under cloud cover, low light, strong sunlight and wind loading is to be verified.

- **Cost & Manufacturability**

Readily available, low-cost components (LM358, BC547/BC557, standard LDRs) are to be preferred and a simple mechanical design is to be adopted to ease replication and repair.

3.5 ERROR REDUCTION TECHNIQUES:

There are many possibilities for an error in a circuit. It can either be due to the error in the component or in the circuit. The common errors and the error reduction techniques to avoid errors are as follows:

1. Matched sensors and resistor tolerances

- Paired LDRs from the same batch are to be used and precision bias resistors (1% tolerance) are to be selected.
- Systematic offset is minimized because both sensors are made to behave identically under equal illumination.
- Similar LDR pairs are to be matched by measurement and 1% metal-film resistors are to be employed in the divider.

2. Symmetrical mechanical placement & shielding

- LDRs are to be placed symmetrically with a small opaque divider and a light-tunnel or hood is to be used to block stray reflections.
- Unequal ambient lighting and reflection-based directional errors are avoided by this arrangement.
- A small tube or slot is to be fabricated (3D-printed or cut) so that only direct light reaches each LDR.

3. Differential amplification with high CMRR

- A differential amplifier stage or op-amp configuration with high Common-Mode Rejection is to be used.
- Variations affecting both sensors equally are rejected, while only the differential component is amplified.
- Op-amps with high CMRR are to be selected and gain-setting resistor pairs are to be matched.

4. Add small hysteresis (dead-band) at comparator

- A small positive feedback path is to be introduced so that a tiny on/off window (dead-band) is created.
- Motor hunting near zero illumination difference is prevented through this hysteresis.
- An appropriate feedback resistor is to be added around the comparator to obtain ~10–50 mV effective hysteresis.

5. Low-pass filtering (RC smoothing) on sensor lines

- Small RC filters are to be added to each sensor node.
- Fast flicker due to passing clouds or shadows is smoothed out.
- A capacitor is to be placed from the sensor node to ground with τ chosen to suppress fast transients but preserve slow sun motion.

6. Mechanical damping & gearing

- A geared motor or mechanical damping is to be employed to slow movement and absorb jerks.
- Overshoot and jitter caused by small signal variations are reduced by this method.
- Gear ratios between 10:1 and 100:1 can be used and frictionless bearings or soft stops may be added.

7. Limit motor activation time & use pulse control

- The motor is to be driven using short pulses or low-duty-cycle PWM for fine corrections.
- Long continuous rotation from minor errors is avoided and energy usage is reduced.
- Brief pulses (200–500 ms) are to be applied per correction before re-evaluating sensor status.

8. Temperature compensation / component selection

- Temperature-induced drift in LDRs and op-amp offsets is to be considered during component selection.
- Slow bias changes over the day are prevented by proper thermal design.
- Low-drift op-amps and thermally stable resistors are to be used; a thermistor can be added if compensation is required.

9. Debounce logic (analog) or simple timing check

- A persistence check (0.2–1 s) is to be required before a direction change is initiated.
- Responses to brief glitches are avoided by this timing requirement.
- RC filtering combined with comparator hysteresis or a monostable timing circuit is to be implemented.

10. Regular cleaning & maintenance schedule

- Sensor housings are to be designed for easy cleaning of dust and soiling.
- Long-term response drift due to soiling is prevented through regular maintenance.
- Small protective covers are to be used and monthly/quarterly cleaning intervals may be recommended.

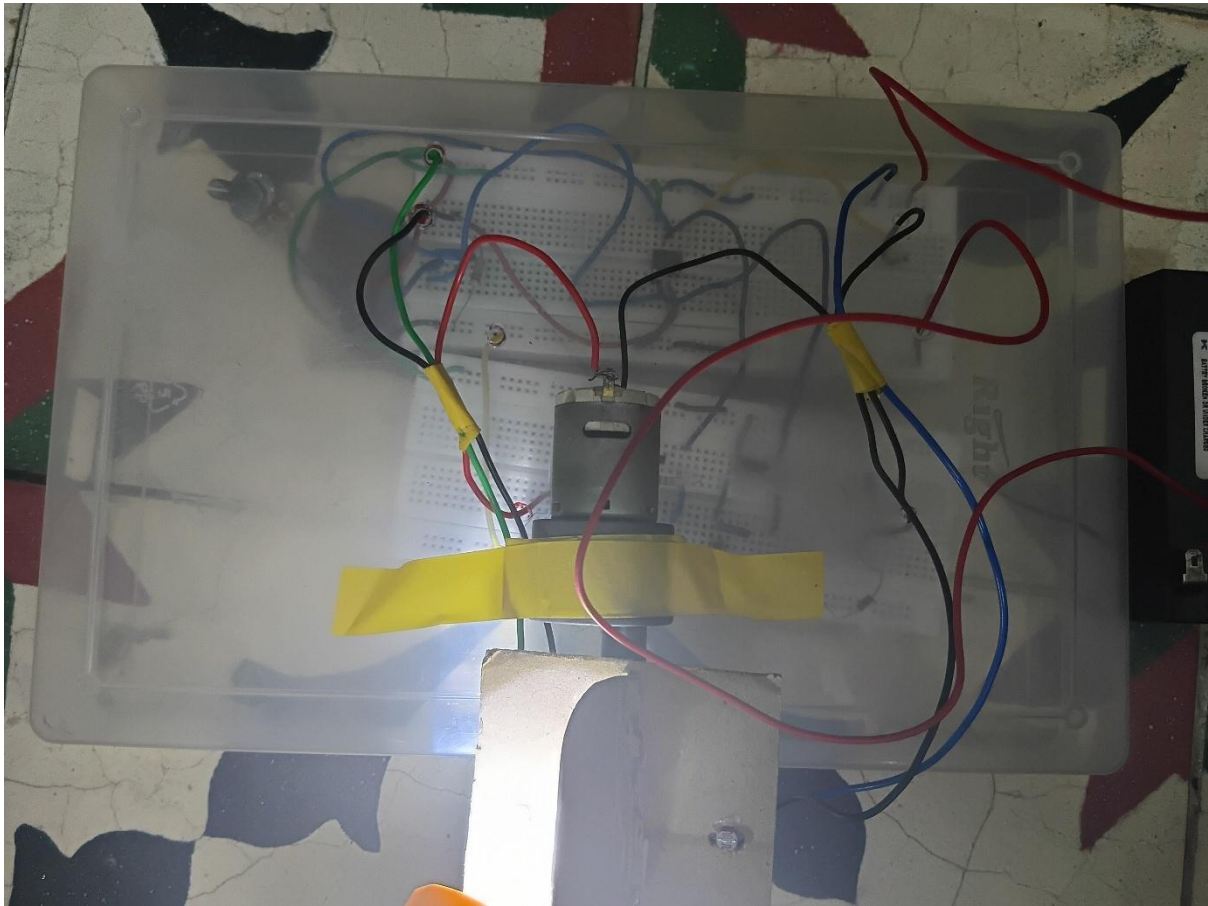


Fig 3.4 Final Output

Overall, the refinements described above ensure that the single-axis solar tracker operates with maximum stability, accuracy and robustness under real-world conditions. By combining sensor matching, mechanical symmetry, filtered differential sensing, controlled motor actuation and periodic calibration, the system achieves reliable sun-tracking performance with minimal energy consumption. These consolidated improvements not only enhance tracking precision but also extend the lifespan of the mechanical and electronic components, resulting in a practical and efficient design suitable for academic, prototype and small-scale solar applications.

CHAPTER 4

CONCLUSION

The Analog Single-Axis Sun Tracker developed in this project successfully demonstrates a simple, low-cost and power-efficient method for maximizing solar energy capture by continuously aligning the solar panel with the direction of maximum sunlight. By using LDR-based light sensing, a differential analog control circuit and a DC motor mechanism, the system is able to autonomously adjust panel orientation without the need for microcontrollers, complex software, or digital processing.

Throughout the design, special focus was given to sensor positioning, noise reduction, stability and incremental motor control to ensure smooth tracking and prevent oscillations. Experimental observations show that even small improvements in panel orientation can significantly enhance the overall energy-harvesting efficiency compared to a fixed panel setup.

This project validates that analog tracking remains a practical and effective solution, especially for small-scale, educational and low-budget renewable energy applications. With further enhancements such as dual-axis extension, low-power motor drivers, or hybrid analog-digital optimization, the system can be adapted for broader and more advanced real-world use cases.

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